

Aditya Wadhwa

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EDUCATION

University of Waterloo

Bachelor of Applied Science, Management Engineering - GPA (4.0)

Waterloo, ON

Sept. 2025 – June 2030

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS, VBA, SQL, Ruby

Frameworks: React, Flask, Ruby on Rails, Llama API

Developer Tools: git/GitHub, Docker, Kubernetes, VS Code, Visual Studio, PyCharm, Power BI

Libraries: pandas, NumPy, Matplotlib, datetime, millify, tkinter, yahooquery

EXPERIENCE

Privy Council Office

Ottawa, ON

Student Data Analyst

May 2026 – Aug. 2026

- Built interactive employment equity dashboards in Tableau and Power BI, enabling directors across the Corporate Services Branch to identify workforce representation gaps and track progress against federal targets.
- Reduced manual preparation time for monthly ministerial regional office tracking reports by approximately 40% by automating recurring data cleaning and report generation workflows in Python and Power Query.
- Improved directors' ability to act on employment equity trends without added analyst support by communicating complex workforce insights to non-technical stakeholders through structured walkthroughs and plain-language summaries.

City of Ottawa

Ottawa, ON

Recreation Assistant / Program Leader

Jan. 2024 – Aug. 2025

- Orchestrated large-scale community events for 1000+ residents by coordinating logistics and ensuring safety compliance across recreational facilities.
- Enhanced operational efficiency by facilitating structured activities for youth groups aged 7–13, ensuring 100% adherence to safety protocols.

PROJECTS

AccessView | JavaScript, OpenLayers, OpenRouteService, Flask, Git

Sept. 2026

- Built an interactive OpenLayers mapping application helping mobility-limited and senior residents discover accessible walking routes and nearby amenities such as pharmacies, clinics, and transit stops, advancing UN SDG 11.2 by ingesting free OpenStreetMap and municipal open-data GeoJSON into styled, category-toggleable vector layers.
- Enabled two-click accessible route planning by integrating the free OpenRouteService API with wheelchair and foot-walking profiles, rendering live turn-by-turn GeoJSON routes and caching repeated requests to remain within free-tier limits at zero API cost.

NASA Space-Apps Hackathon | Python, Three.js, Flask, RandomForest, React

Jan. 2026

- Engineered a high-performance machine learning pipeline aimed at resolving on-going exo-planet detection issues at NASA, RandomForest classifiers on 50GB of NASA Kepler/TESS/K2 datasets.
- Secured a top 4 placement among 150+ participants by pitching a full-stack web application integrated with a Flask backend and Three.js frontend visualizations.

NHL Game Predictor Pipeline | Python, Scikit-learn, Pandas, Git

Jan. 2026

- Developed an end-to-end machine learning pipeline predicting NHL game outcomes by ensembling Logistic Regression and Gradient Boosting classifiers in Scikit-learn, producing calibrated win probabilities that outperform a home-ice baseline (validated via log-loss and Brier score).
- Optimized data ingestion workflows processing 10,000+ match records with minimal latency by engineering automated scraping scripts and feature extraction protocols using Pandas and BeautifulSoup.

Mock Stock Portfolio | Python, YahooQuery API, NumPy

Dec. 2024 – Feb. 2025

- Built a real-time trading simulator enabling risk-free strategy testing by integrating the YahooQuery API with a Python and NumPy backend for high-speed data processing.
- Reduced asset research time by 40% by programming a dynamic stock screener that filters assets based on P/E ratios, ESG scores, and Growth metrics.

Financial Planning Excel Tool | VBA, Macros, Statistical Analysis

Nov. 2025 – Dec. 2025

- Designed an automated budgeting tool improving financial literacy for first-year engineering students by scripting advanced VBA macros and interactive UserForms.
- Streamlined variance analysis by engineering comparison algorithms that visualize Actual vs. Ideal budget differences in real-time.